

Price File Builder

A Windows desktop application that automates the import, transformation, and loading of supplier price files into the RxOne pharmacy database — replacing a time-consuming manual process.

Platform: Windows
Desktop (PyQt6)

Database: Microsoft Access
(suppliers.mdb)

Target Users: Healthsoft
operations staff

Background

THE PROBLEM

- Suppliers send price files in varied CSV and Excel formats with no consistent structure
- Each file required manual column mapping, data cleaning, and re-formatting
- Descriptions had to be manually shortened to fit the 40-character database limit
- Data had to be manually checked against the existing database before loading
- The process was taking **up to an hour per file**

THE SOLUTION

- A guided desktop app with per-supplier memory — settings are saved and reused automatically
- Automatic column detection and named templates for known suppliers
- Intelligent description cleaning, unit normalisation, and bulk truncation tools
- Side-by-side change preview before any data is written to the database
- One-click upsert (insert + update) directly to the Access database



Key Metrics

~1 hr

Previous time per price file

Minutes

Processing time with the app

40 chr

Description limit — automatically enforced

0 SQL

No SQL knowledge required by staff

Main Features



Flexible File Loading

Opens CSV and Excel files (.xlsx / .xls). For multi-sheet workbooks, the user selects which sheet to load. ID-like columns (Pharmacode, Barcode, PLU) are automatically treated as text to prevent formatting errors like "123456.0".



Smart Column Mapping

Maps supplier file columns to standard price file fields: PLU, Pharmacode, Retail, Cost, Barcode/EAN, Supplier Code, and Outer. Pre-built templates for known suppliers auto-populate the mapping; a General template uses fuzzy matching for new suppliers.



Description Builder

Combines multiple columns into a single TradeName field with a configurable separator. Applies title case conversion, unit normalisation (e.g. "75 g" → "75g"), optional brand prefix, and enforces the 40-character database limit with a bulk-truncate tool.



Shortening Rules

Per-supplier find-and-replace rules stored in supplier_rules.json automatically shorten verbose descriptions. Rules support whole-word matching, case-insensitive search, and regular expressions. Rules are editable from within the app.



Supplier Memory

Column mappings, TradeName settings, and skip-row counts are saved per supplier and restored automatically on next use. Up to 6 separate sheet layouts can be saved per supplier for workbooks where different product categories live on different sheets.



Change Preview

Before writing anything, the app compares the processed output against the live Access database. NEW records appear in green; UPDATES appear in amber showing old vs. new description and prices. Rows can be removed from the preview before committing.



Background Prefetch

As soon as a supplier is selected, the app silently fetches their existing database records in the background. This means the "Preview Changes" step is near-instant rather than requiring a separate wait.



Export & Upsert

Output can be exported as a CSV file, or written directly to the Access database with a single click. The upsert matches on Supplier Code first, falling back to Barcode. A "Replace All" option is available for full refreshes of a supplier's records.



Specials File Workflow

A separate mode handles supplier specials promotions. Per-brand column mapping rules are stored in `brand_import_mappings.json` — a plain text file that sits next to the exe and can be edited without a rebuild to add new brands. Descriptions exceeding the 40-character field limit are automatically resolved from the product master. Includes a real-time stock barcode lookup via the Healthsoft SQL Server.

Typical Workflow

1

Select Supplier

Choose the supplier from the dropdown. The app immediately starts fetching their existing records from Access in the background.

2

Open Price File

Browse to the supplier's CSV or Excel file. For multi-sheet workbooks, select the correct sheet.

3

Review Mappings

The template auto-fills column mappings where possible. Review and adjust the column mapping and TradeName settings as needed.

4

Build Output

Click Build Output. The app processes the file, applies TradeName rules, filters invalid rows, and displays the result.

5

Preview & Fix Descriptions

Preview Changes compares the output against the database. Edit descriptions directly in the grid, or use Bulk Truncate to clip everything to 40 characters. Remove any rows not needed.

6

Save Settings & Write to Database

Optionally save the current mappings for this supplier. Then click Execute Upsert to write new and updated records to the Access database — or Download CSV to export the file instead.

Technical Overview

Component	Technology	Notes
Desktop UI	Python / PyQt6	Native Windows application; packaged as a standalone .exe with PyInstaller — no Python install required on end-user machines
Data Processing	pandas	CSV and Excel loading, column transformation, de-duplication, money parsing
Primary Database	Microsoft Access (.mdb) via ODBC	Reads and writes the live suppliers.mdb; supports both local and network (UNC) paths
Fallback Database	SQLite	Available for environments without the Access ODBC driver installed
Stock Lookup	SQL Server via ODBC	Used by the Specials workflow to look up barcodes in the Healthsoft stock database
Configuration	settings.ini	Database paths and credentials stored locally; editable via the app's Settings panel or any text editor
Supplier Templates	JSON files	templates.json, supplier_settings.json, and supplier_rules.json store all per-supplier configuration. brand_import_mappings.json (specials brand column rules) sits next to the exe and is editable without rebuild.
Alternative Interface	Streamlit (web)	A browser-based version of the same workflow is available for environments where the desktop app cannot be installed

